

# Roman F. Rossi

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## EDUCATION

**James Madison University** 2022-2025  
B.S. in Computer Science

**Northern Virginia Community College** 2018-2022

## RELEVANT COURSEWORK

|                                 |                      |
|---------------------------------|----------------------|
| Algorithms & Data Structures    | Cyber Tech Systems   |
| Software Engineering            | Adv Web Tech &       |
| Java Programming                | Multimedia Software  |
| Discrete Structures/Mathematics | IT Database Design & |
| C++ Programming                 | SQL                  |
| Python Programming              | Database Management  |

## EXPERIENCE

### Data Annotation | AI Engineer | Remote | Mar 2025 – Jan 2026

- Evaluated AI model performance across Python, JavaScript, and Rust through code generation testing, building automated test suites to assess correctness and instruction-following capabilities.
- Conducted adversarial red teaming to identify safety vulnerabilities in LLMs, developing targeted prompts to test model robustness in cybersecurity and sensitive use cases.
- Designed evaluation rubrics and prompts for diverse AI tasks including coding, data analysis, and API integration, contributing to model training and assessment frameworks.

### BlueHalo | IT Intern | Arlington, VA | May-Aug 2023

- Managed IT asset inventory across multiple high-security facilities using KACE, tracking hundreds of devices while maintaining compliance with strict security protocols.
- Performed secure data wiping and system reconfiguration for device redeployment and resale, ensuring complete data removal per security standards

## SKILLS

**Programming Languages:** Java, Python, C++, C, SQL, JavaScript, HTML/CSS, Ruby, Rust, Haskell, Batch  
**Technologies & Tools:** Git, Linux, Windows, REST APIs, JSON, YAML, AsyncIO, Multithreading, Docker, KACE, Virtual Machines

**Frameworks & Libraries:** React, Django, Node.js, Swing, NumPy, Pandas, Whisper API, PyAudio, faster-whisper, Anthropic API, Alpaca API

**Development Practices:** Agile/Scrum, Version Control, Code Review, Testing & Debugging, System Architecture, API Integration, Encryption (AES-256)

## PROJECTS

### RCLI – Command-Line Interface Framework | Python, argparse, API Integration

- Architected extensible CLI system managing 50+ utility commands across file processing, media conversion, web scraping, and automation workflows.
- Implemented plugin architecture with dynamic command registration and LLM-powered semantic search for intelligent command discovery.
- Built automation tools integrating YouTube, TikTok, Instagram, and other APIs for content extraction, with support for batch processing.

### AutoTrader – AI-Powered Trading System | Python, AsyncIO, Anthropic API, Alpaca API

- Developed automated trading platform integrating Alpaca API with Claude AI for real-time market analysis and automated trade execution.
- Implemented multi-server MCP architecture for concurrent data aggregation from Yahoo Finance and financial data providers.
- Built risk management system with PDT tracking, dynamic position sizing, and automated safety controls.

### MindLoop – Encrypted Personal Data Manager | Java, Swing, AES Encryption, MVC

- Built Java desktop application with AES-256 encryption for secure personal data tracking and journal management.
- Designed modular architecture with 34 classes across 5 packages (app, gui, other, testing, theme) following MVC pattern.
- Created custom-styled GUI with themed components and real-time encrypted file I/O for data persistence.

### Audio Processing Toolkit | Python, Whisper API, PyAudio, Multithreading

- Developed suite of audio tools including real-time speech-to-text transcription, TTS, and voice conversion using RVC models.
- Implemented faster-whisper with GPU acceleration achieving high-accuracy transcription with real-time microphone input support.
- Built CLI interfaces for batch audio processing with multithreading for concurrent file operations.